

Introduccion a Hugging Face

La plataforma para la comunidad de Machine Learning

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Today's Journey

1. Introduction
2. What is Hugging Face?
3. Hugging Face Ecosystem
4. Start with Hugging Face
 - a. Examples
 - b. Use Cases
5. Resources and links
6. Conclusion

Introduction

If you want to build with models like ChatGPT, BERT or Stable Diffusion...
Hugging Face is the place.

- The biggest repository of open-source models
- More than a million public models
- They have models for NLP, Computer Vision and Reinforcement Learning
- Models in more than 180 languages
- Different libraries of ML could use the HUB

What is Hugging Face

- Company founded in 2016
- A platform to share Machine Learning models
- A global community with a thousands of open-source models and datasets

“The AI community building the future”



The AI community building the future.

The platform where the machine learning community collaborates on models, datasets, and applications.

[Explore AI Apps](#)

or

[Browse 1M+ models](#)

Tasks Libraries Datasets Languages Licenses Other

Filter Tasks by name

Multimodal

- Text-to-Image
- Image-to-Text
- Text-to-Video
- Visual Question Answering
- Document Question Answering
- Graph Machine Learning

Computer Vision

- Depth Estimation
- Image Classification
- Object Detection
- Image Segmentation
- Image-to-Image
- Unconditional Image Generation
- Video Classification
- Zero-Shot Image Classification

Natural Language Processing

- Text Classification
- Token Classification
- Table Question Answering
- Question Answering
- Zero-Shot Classification
- Translation
- Summarization
- Conversational
- Text Generation
- Text2Text Generation
- Sentence Similarity

Audio

- Text-to-Speech
- Automatic Speech Recognition
- Audio-to-Audio
- Audio Classification
- Voice Activity Detection

Tabular

- Tabular Classification
- Tabular Regression

Reinforcement Learning

- Reinforcement Learning
- Robotics

Models 469,541 Filter by name

meta-llama/llama-2-70b

Text Generation • Updated 4 days ago • 25.2k • 64

stabilityai/stable-diffusion-xl-base-0.9

Updated 6 days ago • 2.01k • 393

openchat/openchat

Text Generation • Updated 2 days ago • 1.3k • 136

lillyasviel/ControlNet-v1-1

Updated Apr 26 • 1.87k

cerspense/zeroscope_v2_XL

Updated 3 days ago • 2.66k • 334

meta-llama/llama-2-13b

Text Generation • Updated 4 days ago • 328 • 64

tiiuae/falcon-40b-instruct

Text Generation • Updated 27 days ago • 288k • 899

WizardLM/WizardCoder-150B-V1.0

Text Generation • Updated 3 days ago • 12.5k • 332

CompVis/stable-diffusion-v1-4

Text-to-Image • Updated about 17 hours ago • 448k • 5.72k

stabilityai/stable-diffusion-2-1

Text-to-Image • Updated about 17 hours ago • 782k • 2.81k

Salesforce/xgen-7b-8k-inst

Text Generation • Updated 4 days ago • 6.18k • 57

Hugging Face Ecosystem



Model Hub

- More than 1 million of models
- Samples: `bert-base-uncased` or `stable-diffusion`



Dataset Hub

- Ready to use with `datasets.load_dataset()`



Transformers Library

- Hugging Face's flagship library.
- Allows you to use models like BERT, GPT, T5, etc., with just a few lines of code.



Spaces

- Free hosting for apps built with Gradio or Streamlit.
- Perfect for quick prototyping.

Start with Hugging Face

a. Install the library

```
pip install transformers datasets
```

b. Example

```
from transformers import pipeline

qa = pipeline("question-answering")
result = qa(question="What is Hugging Face?", context="Hugging Face is a company...")
print(result["answer"])
```


Start with Hugging Face

c. Using models of Hugging Face Hub:

```
from transformers import AutoTokenizer, AutoModel

tokenizer = AutoTokenizer.from_pretrained("bert-base-uncased")
model = AutoModel.from_pretrained("bert-base-uncased")
```

Enlace en [github](#):

- https://github.com/GaryBriceno/ml_examples/blob/main/beginners/02_using_hugging_face.ipynb

In Jupyter notebooks:

```
[3]: !pip install transformers datasets
```

```
Collecting transformers
  Downloading transformers-4.53.0-py3-none-any.whl.metadata (39 kB)
Collecting datasets
  Downloading datasets-3.6.0-py3-none-any.whl.metadata (19 kB)
Requirement already satisfied: filelock in /opt/anaconda3/lib/python3.12/site-packages (from transformers) (3.13.1)
Collecting huggingface_hub<1.0,>=0.30.0 (from transformers)
  Downloading huggingface_hub-0.33.2-py3-none-any.whl.metadata (14 kB)
Requirement already satisfied: numpy>=1.17 in /opt/anaconda3/lib/python3.12/site-packages (from transformers) (1.26.4)
Requirement already satisfied: packaging>=20.0 in /opt/anaconda3/lib/python3.12/site-packages (from transformers) (24.1)
Requirement already satisfied: pyyaml>=5.1 in /opt/anaconda3/lib/python3.12/site-packages (from transformers) (6.0.1)
Requirement already satisfied: regex!=2019.12.17 in /opt/anaconda3/lib/python3.12/site-packages (from transformers) (2024.9.11)
Requirement already satisfied: requests in /opt/anaconda3/lib/python3.12/site-packages (from transformers) (2.32.3)
Collecting tokenizers<0.22,>=0.21 (from transformers)
  Downloading tokenizers-0.21.2-cp39-abi3-macosx_11_0_arm64.whl.metadata (6.8 kB)
Collecting safetensors>=0.4.3 (from transformers)
  Using cached safetensors-0.5.3-cp38-abi3-macosx_11_0_arm64.whl.metadata (3.8 kB)
Requirement already satisfied: tqdm>=4.27 in /opt/anaconda3/lib/python3.12/site-packages (from transformers) (4.66.5)
Requirement already satisfied: pyarrow>=15.0.0 in /opt/anaconda3/lib/python3.12/site-packages (from datasets) (16.1.0)
```

Some observations:

- As part of your environment, you need a deep learning library like: PyTorch, TensorFlow or Flax, is necessary to use pre-training models of Hugging Face

```
[5]: from transformers import pipeline
```

```
None of PyTorch, TensorFlow >= 2.0, or Flax have been found. Models won't be available and only tokenizers, configuration and file/data utilities can be used.  
None of PyTorch, TensorFlow >= 2.0, or Flax have been found. Models won't be available and only tokenizers, configuration and file/data utilities can be used.
```

Install the libraries

```
!pip install torch
!pip install transformers datasets
```

```
Requirement already satisfied: torch in /opt/anaconda3/lib/python3.12/site-packages (2.7.1)
Requirement already satisfied: filelock in /opt/anaconda3/lib/python3.12/site-packages (from torch) (3.13.1)
Requirement already satisfied: typing-extensions>=4.10.0 in /opt/anaconda3/lib/python3.12/site-packages (from torch) (4.11.0)
Requirement already satisfied: setuptools in /opt/anaconda3/lib/python3.12/site-packages (from torch) (75.1.0)
Requirement already satisfied: sympy>=1.13.3 in /opt/anaconda3/lib/python3.12/site-packages (from torch) (1.14.0)
Requirement already satisfied: networkx in /opt/anaconda3/lib/python3.12/site-packages (from torch) (3.3)
```

Observations

- In the pipeline **question-answering**, they default model is **distilbert-base-cased-distilled-squad**.
- This is not a best practices, because the model would change in the future.

 Why is this important?

- In PROD, Hugging Face could update or deprecate models.
- If you depend on a model without explicitly specifying its name and version, your result might change unexpectedly.
- It's a good practice to always pin the model and version.

Updating the code:

```
[19]: qa = pipeline(  
    "question-answering",  
    model="distilbert/distilbert-base-cased-distilled-squad",  
    framework="pt" # <- Forzar uso de PyTorch  
)  
  
result = qa(  
    question="What is Hugging Face?",  
    context="Hugging Face is a company."  
)  
  
print("Respuesta:", result["answer"])
```

Device set to use mps:0

Respuesta: a company

“En Mac, Hugging Face can run models using MPS, Apple’s engine, to accelerate AI models.”

Use case

- Custom chatbots
- Sentiment analysis
- Machine translation
- Text generation (GPT, T5)
- Computer vision with CLIP or Stable Diffusion

Resources

- Web :
 - <https://huggingface.co>
- Hugging Face Course:
 - <https://huggingface.co/course>
- Huggin Face Repo:
 - <https://github.com/huggingface>
- Huggin Face Transformers:
 - <https://github.com/huggingface/transformers>

Conclusion

- Hugging Face democratizes access to powerful AI
- The ecosystem is built for collaboration and speed
 - Hugging Faces accelerates the prototyping and production
- Whether you're doing NLP, vision or multimodal work, Hugging Face has a tool for you.

Gracias